# MPG No. 8.4.27 $m'm'm$ (m'm'm setting) [ Type III, orthorhombic ] [G tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{yzx} & 0 \\ 0 & 0 & G_{yzx} \end{bmatrix}$$

$$G_{yzx} = \mathbb{G}^{(1)}_{xyz}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$G_{yzx} = 3\mathbb{Q}^{(1)}_u + \mathbb{Q}^{(1)}_v$$
$$G_{zxy} = -3\mathbb{Q}^{(1)}_u + \mathbb{Q}^{(1)}_v$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = \mathbb{Q}^{(1)}_0 - \mathbb{Q}^{(2)}_u + \mathbb{Q}^{(2)}_v$$
$$G_{zxy} = \mathbb{Q}^{(1)}_0 - \mathbb{Q}^{(2)}_u - \mathbb{Q}^{(2)}_v$$
$$G_{xyz} = \mathbb{Q}^{(1)}_0 + 2\mathbb{Q}^{(2)}_u$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).